Table 3. Screen and supporting analyses — A

BBJ discovery gene	BBJ OR (95% CI)	BBJ P	FinnGen GD OR (95% CI)	FinnGen GD P	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	**2.60×10^{-23}**	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	**8.30×10^{-17}**	1.90 (1.64–2.20)	**1.41×10^{-17}**	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.17 (0.08–0.36)	**2.15×10^{-6}**	Yes	0.254
TSHR	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.08 (0.05–0.12)	**3.19×10^{-32}**	Yes	0.264
C4A	0.45 (0.35–0.57)	**2.23×10^{-10}**	0.41 (0.26–0.63)	**4.97×10^{-5}**	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	**2.04×10^{-7}**	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	**3.14×10^{-7}**	3.28 (2.09–5.17)	**2.68×10^{-7}**	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	**6.35×10^{-7}**	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	**1.48×10^{-6}**	0.77 (0.65–0.92)	**0.00361**	Yes	0.303
PRSS36	6.00 (2.87–12.57)	**2.02×10^{-6}**	1.40 (1.00–1.96)	**0.049**	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	**5.28×10^{-6}**	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	**6.86×10^{-6}**	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	**9.41×10^{-6}**	1.62 (1.18–2.24)	**0.0033**	Yes	0.313

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Table 3. Screen and supporting analyses — B

Additional gene	GD IVs	GD OR (95% CI)	GD P	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	**7.52×10^{-12}**	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	**7.52×10^{-12}**	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	**4.65×10^{-10}**	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	**1.51×10^{-9}**	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	**3.75×10^{-8}**	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	**4.89×10^{-7}**	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	<0.001	<0.001	0.311

nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. APOM/NEU1 share rs3101018; IGLV2-18/IGLV3-12 share rs5759410. The lead IGLV expression variant is absent in BBJ, weakening interpretation of its low PP.H4.

Table 3. Screen and supporting analyses — C

Gene / GD analysis	Reference OR (95% CI)	Reference P	Cohort-frequency OR (95% CI)	Cohort P	Interpretation
C4B	1.63 (1.31–2.03)	**1.10×10^{-5}**	1.75 (1.31–2.33)	**1.48×10^{-4}**	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	**1.72×10^{-5}**	Weight-sensitive; single-variant dependence

HLA-G was added because of frequency sensitivity, not primary discovery. Neither C4B nor HLA-G gains strong GD colocalization. All 24 genes and five datasets are available in the workbook. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.